

```
import os
print(os.getcwd())
```

```
/content
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True)
```

```
!nvidia-smi
```

Wed Jul 1 07:30:57 2026

NVIDIA-SMI 580.82.07				Driver Version: 580.82.07				CUDA Version: 13.0			
GPU		Name	Persistence-M	Bus-Id	Disp.A	Volatile	Uncorr.	ECC			
Fan	Temp	Perf	Pwr:Usage/Cap		Memory-Usage	GPU-Util	Compute	M.			
							MIG	M.			
0	Tesla T4		Off	00000000:00:04:0	Off			0			
N/A	49C	P8	9W / 70W		0MiB / 15360MiB	0%	Default	N/A			

Processes:

GPU	GI	CI	PID	Type	Process name	GPU Memory
	ID	ID				Usage

No running processes found

```
!wget https://bitbucket.org/ishaanjav/code-and-deploy-custom-tensooodel/raw/a4febbfee178324b2083e322cdead7465d6fdf95/fruits.
--2026-07-01 07:31:02-- https://bitbucket.org/ishaanjav/code-and-deploy-custom-tensooodel/raw/a4febbfee178324b2083e322cdead7
Resolving bitbucket.org (bitbucket.org)... 13.200.41.134, 13.200.41.135, 13.200.41.136, ...
Connecting to bitbucket.org (bitbucket.org)|13.200.41.134|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-07-01 07:31:02 ERROR 404: Not Found.
```

```
!ls /content
```

```
drive fruits.zip sample_data
```

```
!file fruits.zip
```

```
fruits.zip: Zip archive data, at least v2.0 to extract, compression method=store
```

```
!unzip fruits.zip
```

```

inflating: fruits/validation/orange/Screen Shot 2018-06-12 at 11.58.11 PM.png
inflating: __MACOSX/fruits/validation/orange/._Screen Shot 2018-06-12 at 11.58.11 PM.png
inflating: fruits/validation/orange/Screen Shot 2018-06-12 at 11.56.55 PM.png
inflating: __MACOSX/fruits/validation/orange/._Screen Shot 2018-06-12 at 11.56.55 PM.png
inflating: fruits/validation/orange/Screen Shot 2018-06-12 at 11.58.50 PM.png
inflating: __MACOSX/fruits/validation/orange/._Screen Shot 2018-06-12 at 11.58.50 PM.png
inflating: fruits/validation/orange/Screen Shot 2018-06-12 at 11.56.02 PM.png
inflating: __MACOSX/fruits/validation/orange/._Screen Shot 2018-06-12 at 11.56.02 PM.png
inflating: fruits/validation/orange/Screen Shot 2018-06-12 at 11.58.24 PM.png
inflating: __MACOSX/fruits/validation/orange/._Screen Shot 2018-06-12 at 11.58.24 PM.png
inflating: fruits/validation/orange/Screen Shot 2018-06-12 at 11.58.28 PM.png
inflating: __MACOSX/fruits/validation/orange/._Screen Shot 2018-06-12 at 11.58.28 PM.png
inflating: fruits/validation/orange/Screen Shot 2018-06-12 at 11.57.37 PM.png
inflating: __MACOSX/fruits/validation/orange/._Screen Shot 2018-06-12 at 11.57.37 PM.png
inflating: fruits/validation/orange/Screen Shot 2018-06-12 at 11.58.02 PM.png
inflating: __MACOSX/fruits/validation/orange/._Screen Shot 2018-06-12 at 11.58.02 PM.png
inflating: fruits/validation/orange/Screen Shot 2018-06-12 at 11.57.52 PM.png
inflating: __MACOSX/fruits/validation/orange/._Screen Shot 2018-06-12 at 11.57.52 PM.png
inflating: fruits/validation/orange/Screen Shot 2018-06-12 at 11.57.07 PM.png
inflating: __MACOSX/fruits/validation/orange/._Screen Shot 2018-06-12 at 11.57.07 PM.png
inflating: fruits/validation/orange/Screen Shot 2018-06-12 at 11.58.18 PM.png
inflating: __MACOSX/fruits/validation/orange/._Screen Shot 2018-06-12 at 11.58.18 PM.png
inflating: fruits/validation/orange/Screen Shot 2018-06-12 at 11.56.48 PM.png
inflating: __MACOSX/fruits/validation/orange/._Screen Shot 2018-06-12 at 11.56.48 PM.png
inflating: fruits/validation/orange/Screen Shot 2018-06-12 at 11.56.35 PM.png
inflating: __MACOSX/fruits/validation/orange/._Screen Shot 2018-06-12 at 11.56.35 PM.png
inflating: fruits/validation/orange/Screen Shot 2018-06-12 at 11.58.43 PM.png
inflating: __MACOSX/fruits/validation/orange/._Screen Shot 2018-06-12 at 11.58.43 PM.png

```

```

import tensorflow as tf
import matplotlib.pyplot as plt

```

```
tf.__version__
```

```
'2.20.0'
```

```

img_height, img_width = 32 , 32
batch_size = 20

train_ds = tf.keras.utils.image_dataset_from_directory(
    "fruits/train",
    image_size = (img_height, img_width),
    batch_size = batch_size
)
val_ds = tf.keras.utils.image_dataset_from_directory(
    "fruits/validation",
    image_size = (img_height, img_width),
    batch_size = batch_size
)
test_ds = tf.keras.utils.image_dataset_from_directory(
    "fruits/test",
    image_size = (img_height, img_width),
    batch_size = batch_size
)

```

```

Found 460 files belonging to 3 classes.
Found 66 files belonging to 3 classes.
Found 130 files belonging to 3 classes.

```

```

import matplotlib.pyplot as plt

class_names = ["apple", "banana", "orange"]

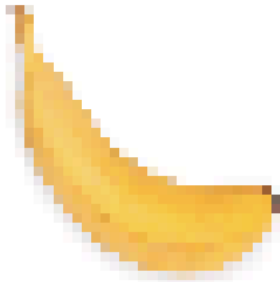
plt.figure(figsize=(10, 10))

for images, labels in train_ds.take(1):
    for i in range(9):
        ax = plt.subplot(3, 3, i + 1)
        plt.imshow(images[i].numpy().astype("uint8"))
        plt.title(class_names[labels[i]])
        plt.axis("off")

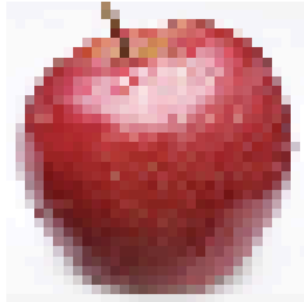
plt.show()

```

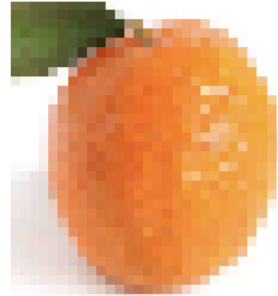
banana



apple



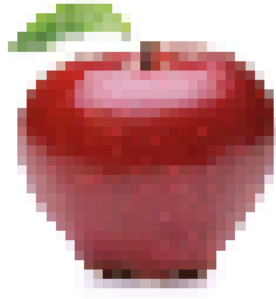
orange



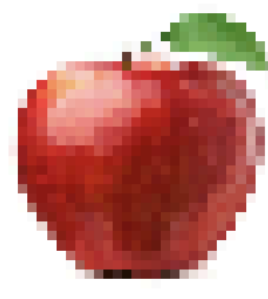
apple



apple



apple



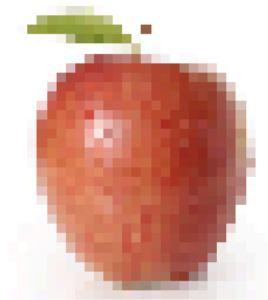
apple



banana



apple



```
model = tf.keras.Sequential([
    tf.keras.layers.Rescaling(1./255),
    tf.keras.layers.Conv2D(32, 3, activation="relu"),
    tf.keras.layers.MaxPooling2D(),
    tf.keras.layers.Conv2D(32, 3, activation="relu"),
    tf.keras.layers.MaxPooling2D(),
    tf.keras.layers.Conv2D(32, 3, activation="relu"),
    tf.keras.layers.MaxPooling2D(),
    tf.keras.layers.Flatten(),
    tf.keras.layers.Dense(128, activation="relu"),
    tf.keras.layers.Dense(3)
])
```

```
model.compile(
    optimizer="rmsprop",
    loss=tf.keras.losses.SparseCategoricalCrossentropy(from_logits=True),
    metrics=["accuracy"]
)
```

```
model.fit(
    train_ds,
    validation_data=val_ds,
    epochs=20
)
```

[Show hidden output](#)

```
import numpy

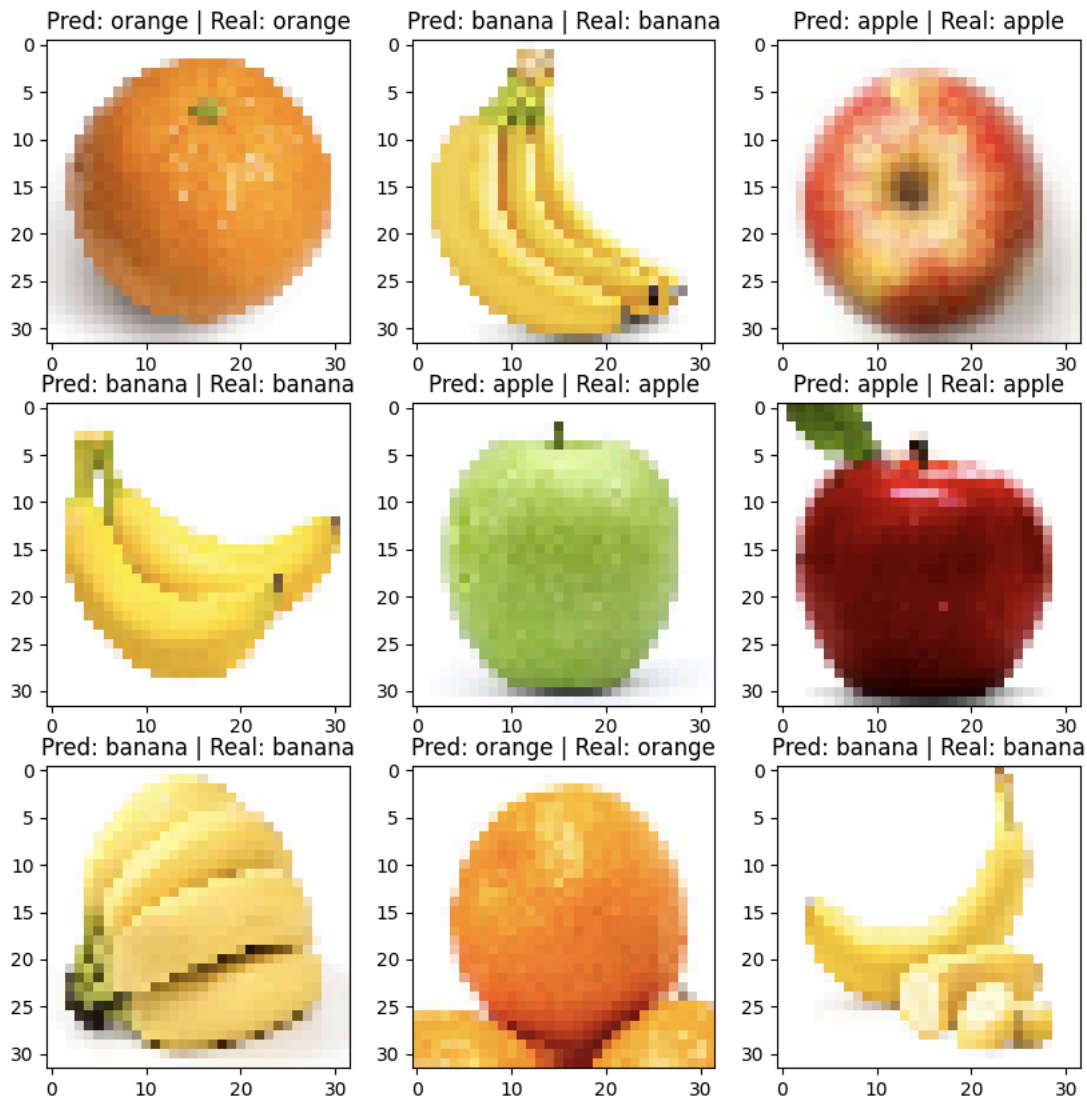
plt.figure(figsize=(10,10))
for image, labels in test_ds.take(1):
    classifications = model.predict(image)
    #print classification
```

```

for i in range(9):
    ax = plt.subplot(3,3,i+1)
    plt.imshow(image[i].numpy().astype("uint8"))
    index = numpy.argmax(classifications[i])
    plt.title("Pred: " + class_names[index] + " | Real: " + class_names[labels[i]])

```

1/1 ————— 0s 273ms/step



```

converter = tf.lite.TFLiteConverter.from_keras_model(model)
tflite_model = converter.convert()

```

```

with open("model.tflite", 'wb') as f:
    f.write(tflite_model)

```

Saved artifact at '/tmp/tmpyqxppd1v'. The following endpoints are available:

```

* Endpoint 'serve'
  args_0 (POSITIONAL_ONLY): TensorSpec(shape=(None, 32, 32, 3), dtype=tf.float32, name='keras_tensor')
Output Type:
  TensorSpec(shape=(None, 3), dtype=tf.float32, name=None)
Captures:
138937829153616: TensorSpec(shape=(), dtype=tf.resource, name=None)
138937829157072: TensorSpec(shape=(), dtype=tf.resource, name=None)
138937829157264: TensorSpec(shape=(), dtype=tf.resource, name=None)
138937829158224: TensorSpec(shape=(), dtype=tf.resource, name=None)
138937829157840: TensorSpec(shape=(), dtype=tf.resource, name=None)
138937829158800: TensorSpec(shape=(), dtype=tf.resource, name=None)
138937829157648: TensorSpec(shape=(), dtype=tf.resource, name=None)
138937829159760: TensorSpec(shape=(), dtype=tf.resource, name=None)
138937937686224: TensorSpec(shape=(), dtype=tf.resource, name=None)
138937829160720: TensorSpec(shape=(), dtype=tf.resource, name=None)

```

Start coding or [generate](#) with AI.

